AI Tools for Managing Infodemics

Health Misinformation in Video-based Social Media

Spring 2026 End of Semester Report

Course: AI Tools for Managing Infodemics (Health Misinformation in Video-based Social Media)
University of Illinois Urbana-Champaign | CHIME Lab

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Abstract

Health misinformation in short-form video has become one of the more pressing challenges in digital public health. This report documents the design, implementation, and evaluation of the WHO Infodemic Monitor, a cloud-deployed AI platform that automatically detects, classifies, and explains health misinformation in TikTok, Instagram Reels, and YouTube Shorts videos. The system combines audio transcription, large language model reasoning, and PubMed evidence grounding to produce structured, explainable verdicts for public health researchers. Three student research tracks ran alongside the production system. The first investigated tradeoffs between audio-only, visual-only, and multimodal classification pipelines. The second analyzed agreement and disagreement patterns across multiple open-weight LLMs. The third explored claim-level classification as a more transparent alternative to video-level labels. Together, these contributions form a cohesive research and engineering effort aligned with WHO's stated priorities for AI-assisted infodemic management.

1. Introduction

The World Health Organization defines an infodemic as too much information including false or misleading information in digital and physical environments during a disease outbreak (WHO, 2024). While infodemics have always existed, the rise of short-form video platforms has fundamentally changed their scale and speed. TikTok reached 1.59 billion monthly active users in early 2025, with users spending an average of 95 minutes per day on the platform. YouTube Shorts surpassed 2 billion logged-in monthly viewers, and Instagram Reels consistently outperforms other content formats in reach and engagement. Together, these three platforms represent an unprecedented distribution channel for health content, accurate and otherwise.

Empirical audits of TikTok health content paint a consistent and concerning picture. Studies on sinusitis found roughly 44% of videos contained non-factual information. A separate study found that over 20% of sexual health-related TikToks were inaccurate. A 2026 systematic review of 27 studies found that TikTok leads all social media platforms in mental health misinformation, with ADHD misinformation rates of 52% and autism misinformation rates of 41%, compared to approximately 22% on YouTube. These are not isolated findings. They converge across topic areas and research groups, pointing to a structural, platform-level problem rather than individual bad actors.

WHO and TikTok announced a formal collaboration in September 2024, with WHO's Fides network of over 800 science communicators reaching 150 million people involved in the effort. This institutional recognition of the problem underscores the urgency of scalable, automated solutions. Manual review of short-form health video is simply not feasible at the volume these platforms generate.

This report documents our semester-long effort to build and evaluate exactly such a solution. The WHO Infodemic Monitor is a production-grade, cloud-deployed AI platform designed for WHO representatives and public health researchers. It ingests a short-form video, transcribes the audio, classifies the content using a large language model, grounds the verdict in peer-reviewed biomedical literature, and returns a structured, explainable result in under 10 seconds. Three research teams investigated key questions about pipeline design, model reliability, and classification granularity, producing findings that directly inform how the system should evolve.

2. Related Work

2.1 Health Misinformation Detection in Short-Form Video

Automated misinformation detection in video is a relatively recent research area. Early work focused on text and images; short-form video introduces the additional challenge of combining spoken content, on-screen text, and visual context in a compressed format. Recent work combining text with OCR-extracted visual features and social context improved classification F1 by roughly 15% over unimodal baselines (Shahi et al., 2025). Liu et al. (2024) found that audio features significantly improve detection performance, with wav2vec-based encoders outperforming visual feature extractors because they align more closely with the semantic representations used by LLMs. Critically, poorly aligned visual features can actively degrade accuracy, which motivates an audio-first design rather than always-on multimodal fusion. ViMGuard (Choi et al., 2024) proposed a two-stage claim detection and verification pipeline for video misinformation, noting that LLMs struggle to verify claims within the complex contextual structure of a full video without first decomposing it.

2.2 LLM Ensembles for Classification Reliability

Single LLMs exhibit meaningful variance on classification tasks, particularly in sensitive domains. Pelrine et al. (2024) compared Llama, Mistral, Falcon, and Orca on COVID-19 misinformation datasets and found substantial accuracy differences, attributing the gaps to RLHF safety tuning. Each model's guardrails create systematic blind spots, making model agreement analysis informative about case difficulty. Farajtabar et al. (2024) showed that LLM chain ensembles reduced performance standard deviation by nearly three-fold and outperformed the best single model by 2.23 F1 points on misinformation detection tasks. A 2025 multi-LLM study measuring inter-model Cohen's kappa found that frontier models achieved values in the almost perfect agreement band above 0.80, suggesting that model agreement is a meaningful signal for production routing decisions.

2.3 PubMed Grounding for Health Claim Verification

Grounding AI outputs in peer-reviewed biomedical literature addresses a core limitation of standalone LLMs: hallucination on health-specific facts. MEGA-RAG (2025) combined PubMed, WHO IRIS, and biomedical knowledge graph retrieval to reduce hallucination rates by over 40% compared to standalone LLM baselines. MedRAGChecker (2026) introduced claim-level NLI verification over biomedical RAG systems, demonstrating that verifying individual extracted claims against retrieved literature produces more faithful and auditable outputs than verifying an entire transcript holistically. Kotonya and Toni (2020) established through the PUBHEALTH benchmark that explainable health fact-checking, where verdicts are linked to specific evidence, produces substantially better trust and usability outcomes than opaque classification scores.

2.4 Claim-Level vs. Video-Level Classification

Binary video-level labels applied to an entire video are increasingly recognized as insufficient for real-world fact-checking. Work on misinformation span detection argues that video-level labels do not indicate when within a video misinformation occurs or which specific claims are responsible (ArXiv, 2026). ViClaim (2025) introduced a multilingual multilabel dataset specifically for in-video claim detection. Document-level claim extraction research has established that decontextualized atomic claims yield more verifiable and actionable outputs than holistic document labels. These findings motivate the claim-level classification track explored in Section 6.

3. The WHO Infodemic Monitor: Production System

3.1 Motivation and Design Goals

The WHO Infodemic Monitor was built with three primary goals. First, it needed to be accessible to non-technical users, specifically WHO researchers and public health analysts rather than software engineers. Second, it needed to produce not just a label but an explanation, a natural language account of why the content was flagged, with specific evidence from the video itself. Third, it needed to be grounded in peer-reviewed literature for credibility in a public health context, where unsupported AI verdicts carry little weight.

These goals shaped every architectural decision: the choice of a web-based frontend, the use of structured LLM output with explicit reasoning, and the PubMed grounding step that runs for MISINFO and DEBUNKING results.

3.2 Classification Schema

The system classifies each video into one of four labels:

Label	Meaning
MISINFO	The video contains health misinformation
NO_MISINFO	The content is accurate or neutral
DEBUNKING	The video actively refutes a health misinformation claim
CANNOT_RECOGNIZE	Insufficient signal to classify reliably

The four-class schema extends the binary MISINFO/NO_MISINFO framing common in the literature. The DEBUNKING label is particularly important: a video that debunks a myth is not misinformation and should not be flagged as such, yet it contains health-related claims that warrant evidence grounding.

3.3 Analysis Pipeline

Figure 1 shows the five-stage pipeline that processes each submitted video.

Figure 1. Analysis pipeline from video upload through structured result. PubMed grounding runs only for MISINFO and DEBUNKING results.

A user uploads a video through the web interface. The audio track is extracted and transcribed using OpenAI's Whisper-1 API. The transcript is passed to Anthropic's Claude with a structured classification prompt that asks the model to assign one of the four labels, provide a confidence score between 0 and 1, generate a natural language explanation, and extract specific evidence snippets from the transcript. For MISINFO and DEBUNKING results, the system then runs a PubMed grounding step: the LLM extracts 2 to 3 specific health claims from the transcript, queries the PubMed E-utilities API for each claim in parallel, and returns linked peer-reviewed citations alongside the verdict.

3.4 Cloud Architecture

Figure 2 shows the four managed cloud services that host the system.

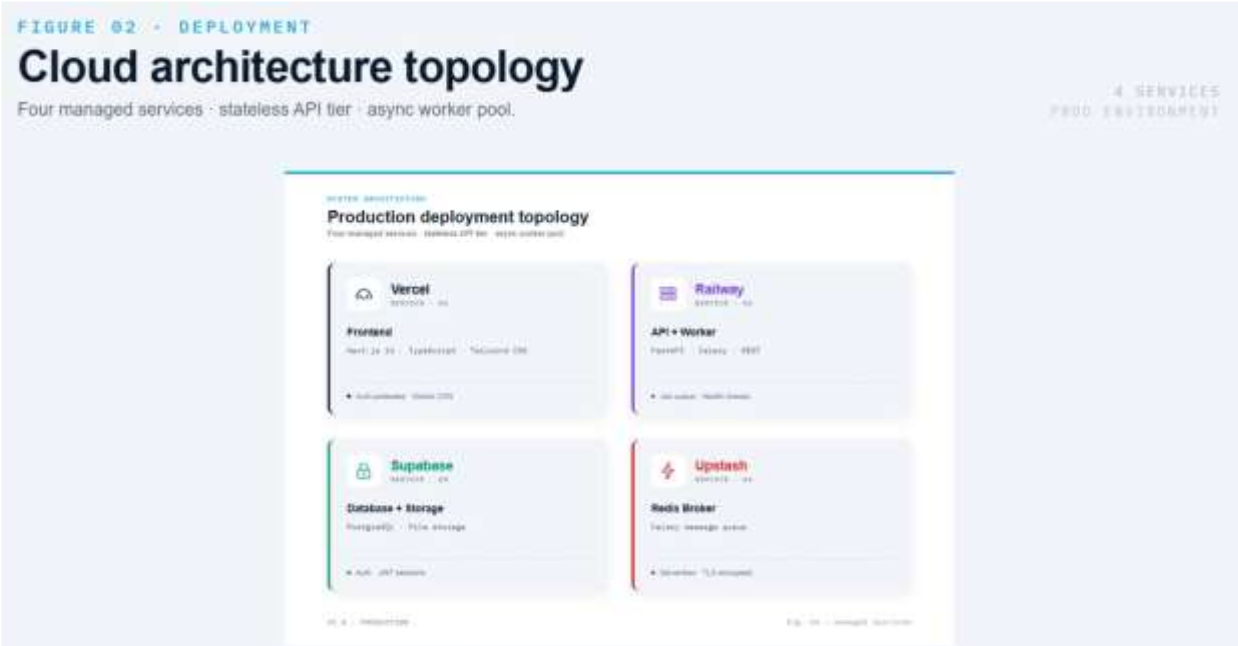

Figure 2. Production deployment topology across Vercel, Railway, Supabase, and Upstash.

The frontend runs on Vercel, providing a globally distributed Next.js application with Supabase authentication protecting all routes. The backend API runs on Railway, hosting a FastAPI server that handles video uploads, job creation, and result retrieval. Video files and analysis results are stored on Supabase, which also manages user authentication through JWT-based sessions. The Celery task queue communicates through an Upstash Redis broker, providing reliable message passing between the API and the analysis worker.

This architecture is fully cloud-hosted, authentication-protected, and horizontally scalable. The inference provider and transcription provider are both controlled by environment variables, allowing the system to adapt to cost or availability constraints without code changes.

3.5 Technology Stack

Layer	Technology
Frontend	Next.js 14, TypeScript, Tailwind CSS
Backend API	FastAPI (Python)
Task queue	Celery + Upstash Redis
Database	Supabase PostgreSQL
File storage	Supabase Storage
Authentication	Supabase Auth + PyJWT
Transcription	OpenAI Whisper-1 / faster-whisper
LLM inference	Anthropic Claude / OpenAI GPT
Evidence grounding	PubMed E-utilities API

Migrations	Alembic + SQLAlchemy
Deployment	Vercel (frontend), Railway (backend)

3.6 Live Results

The system was validated on four videos spanning all four classification categories. Results were produced end-to-end in production through the live web interface.

Video	Label	Confidence	Avg. Latency
Weight loss claims (15 lbs in 20 days)	MISINFO	85%	~6 seconds
Physics demonstration (ice melting)	NO_MISINFO	85%	~6 seconds
Giants myth debunked	DEBUNKING	92%	~6 seconds
Electrolytes hydration tip	NO_MISINFO	95%	~6 seconds

For the MISINFO result, the system generated a detailed explanation citing the CDC's recommended safe weight-loss rate of 1 to 2 pounds per week, identified the specific false metabolic claim, and returned PubMed citations on glucose metabolism and fluid balance. For the DEBUNKING result, the system correctly identified that the video was refuting a false claim rather than making one, and grounded its verdict in papers on Carboniferous-period atmospheric oxygen and arthropod physiology. These results demonstrate not just correct labeling but nuanced, evidence-anchored reasoning of the kind required for public health decision support.

4. Task 3: Modality Comparison Study

Aanchal Malhotra, Soobin Jang

4.1 Overview

A key design question in any multimodal misinformation detection system is whether combining audio transcription and visual OCR actually improves classification performance over audio alone. This question has direct cost implications: OCR processing adds latency and compute. If it provides no accuracy benefit for short-form health videos, then a simpler audio-first pipeline is the correct production choice.

4.2 Methodology

Five TikTok videos were collected covering a range of health content, including both factual information and misinformation. Three processing modes were applied to each video:

- Audio-only: Whisper transcribes the speech, and the transcript is passed to Mistral 7B for classification

- Visual-only: EasyOCR extracts on-screen text from sampled video frames, and the extracted text is passed to Mistral 7B
- Multimodal: Both transcript and OCR text are combined and passed to Mistral 7B together

All other variables were held constant: the same Mistral 7B model, the same classification prompt, the same inference parameters, and the same preprocessing pipeline. Only the input modality changed, ensuring that any differences in output could be attributed to the modality rather than confounds.

4.3 Results

Figure 3 shows the per-video labels and confidence scores across all three modes.

Figure 3. Labels and confidence scores across audio, visual, and multimodal modes for all five test videos. All three modes produced identical outputs.

The results were unambiguous. Across all five videos and all three processing modes, the predicted labels were identical and the confidence scores were identical, averaging 0.77 across all modes. The percentage of cases where adding OCR changed the classification was 0.00. Processing latency showed a slight advantage for the visual-only mode (11.82 seconds average), with audio-only at 12.69 seconds and multimodal at 12.32 seconds.

4.4 Analysis

The most significant finding is that audio and visual modalities in the selected videos contained largely redundant information. Both channels conveyed similar semantic signals, and combining them did not introduce new classification-relevant content. This is consistent with Liu et al. (2024), who found that audio features align more closely with LLM text processing and that poorly aligned visual features can actively degrade accuracy.

The two videos classified as CANNOT_RECOGNIZE across all three modes scored 0.50 confidence in every mode. These videos lacked sufficient signal in both audio and visual channels simultaneously. Multimodal processing is theoretically most valuable when one modality is strong and the other is weak. The absence of improvement here indicates both modalities failed simultaneously, pointing to a data quality issue rather than a modality limitation.

4.5 Production Recommendation

Based on these findings, the recommended production pipeline is:

- Run audio-only transcription by default, as it is sufficient for confident classification in the majority of cases
- Trigger OCR only when audio confidence falls below 0.60, targeting cases where audio is genuinely weak
- Skip multimodal processing for high-confidence results, as there is no accuracy benefit and additional cost is incurred

This conditional design provides the same classification accuracy as always-on multimodal processing at significantly lower compute cost.

4.6 Limitations

The primary limitation is the small dataset size of five videos. A larger and more diverse dataset is needed to confirm that the 0% label-change finding holds across different content types, particularly videos where audio and visual channels carry genuinely different information.

5. Task 4: Model Disagreement Analysis

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5.1 Overview

A separate but complementary research question concerns the reliability of individual LLM classifiers. When multiple models are run on the same video, do they agree? And what does disagreement tell us about classification confidence and production reliability? This track analyzed three open-weight LLMs, Llama 2, Mistral, and Gemma, across a set of TikTok health videos, producing model agreement and disagreement matrices that quantify inter-model consistency.

5.2 Methodology and Assumptions

New TikTok videos were collected covering both factual health content and misinformation. Videos were preprocessed using the same Whisper transcription and OCR pipeline as the main system. Each of the three models processed the exact same inputs under controlled conditions: identical prompt formatting, identical inference parameters, and identical preprocessing. The analysis was conducted under the assumption that confidence values correlate with prediction certainty and that agreement and disagreement patterns are not purely random but can reveal systematic weaknesses in model behavior.

5.3 Results

Figure 4 shows the model agreement and disagreement matrices for Llama 2 and Mistral.

Figure 4. Pairwise agreement matrix (left, all 1.000) and disagreement matrix (right, all 0.000) for Llama 2 and Mistral across all test videos.

Across all test videos, Llama 2 and Mistral produced identical classification labels, with a model agreement score of 1.000 across every case and 0.000 on the disagreement matrix. All three models, including Gemma, were in full agreement for all test cases.

5.4 Analysis

Full model agreement on a classification task can be interpreted in two ways. The optimistic reading is that the test cases were clear and unambiguous, with all three models converging on the same verdict. The more cautious reading, which the team appropriately flags, is that the test cases may have been predominantly straightforward examples without significant edge cases. If the dataset were expanded to include more ambiguous or adversarial health claims, the models might begin to diverge.

Pelrine et al. (2024) documented that these same model families show accuracy gaps on COVID-19 misinformation because each model's RLHF safety tuning creates systematic blind spots. Full agreement on a small, non-adversarial dataset should therefore be understood as a baseline rather than a production guarantee.

The production value of this work is concrete regardless. Full agreement between models can serve as a routing signal: cases where all models agree at high confidence can be routed to automated output, while cases where models disagree can be flagged for human review or secondary verification.

5.5 Limitations

The primary limitation is dataset size and diversity. The agreement matrices are based on a limited number of test videos, and the lack of adversarial examples means we cannot characterize model behavior in cases where automated systems are most likely to fail. Future work should test specifically on cases where individual models are known to disagree, such as claims about vaccine safety or contested nutritional science.

6. Task 5: Claim-Level Classification

Upasana, Suha, Ananya

6.1 Overview

The production system and research tracks described above operate at the video level: a single label and explanation is assigned to the entire video. This has a fundamental limitation. A single video may contain both accurate and inaccurate health claims, and a video-level label cannot distinguish between them. A video that is 90% accurate but contains one dangerous false claim will receive the same MISINFO label as a video that is entirely fabricated.

Claim-level classification addresses this by decomposing the video into individual health assertions, classifying each one separately, and grounding each verdict in specific evidence. The output is more transparent, auditable, and actionable: a public health researcher can see not just that a video was flagged, but exactly which statements triggered the flag and why.

6.2 Approach

Using the existing multimodal pipeline, the team tested how large language models identified explicit health-related claims from short-form video transcripts and structured them into JSON outputs containing the claim text, a classification label for that specific claim, and a confidence score. The workflow extended the existing system's prompt structure to request explicit claim decomposition before classification, allowing the same Claude model used for video-level classification to produce per-claim verdicts.

The team evaluated these outputs for consistency (whether the same claims are identified across runs), interpretability (whether the extracted claims are readable and specific), and alignment with the video-level verdict (whether the claim-level labels collectively support the overall classification).

6.3 Findings

The exploratory testing found that claim-level outputs meaningfully improved explainability by linking misinformation decisions directly to individual statements within a video. Rather than a single explanation paragraph, the system produced a structured breakdown of each health assertion, its classification, and supporting evidence. The outputs were generally interpretable and consistent, with the LLM reliably identifying the core health claims in each video and assigning confidence scores that aligned with the video-level verdict.

Formal evaluation against a labeled ground truth was not completed within the semester. This work should be understood as a feasibility study and workflow validation rather than a performance evaluation.

6.4 Future Direction

Claim-level classification is well positioned for continued development. The next steps would include building a labeled evaluation dataset of individual health claims using tools such as ClaimBuster or a PUBHEALTH subset as reference, integrating claim-level outputs with the PubMed grounding step so each extracted claim generates its own citation retrieval query, and surfacing claim-level breakdowns in the user interface alongside the video-level verdict.

7. Discussion

7.1 What the Research Tracks Tell Us Together

The three research tracks are not independent. They converge on a coherent set of design principles for the next version of the system.

Task 3's finding that audio-only classification is sufficient for the majority of short-form health videos supports the production system's current architecture. The conditional OCR recommendation gives the system a principled mechanism for handling the minority of videos where audio is insufficient, without paying the OCR cost on every video.

Task 4's finding that Llama 2, Mistral, and Gemma agree on clear cases validates the production system's use of a single LLM for primary classification while pointing toward ensemble routing as a reliability mechanism. When the system encounters a low-confidence case, running multiple models and routing based on agreement provides a more robust verdict than simply lowering a confidence threshold.

Task 5's claim-level exploration points toward a richer output format. Moving to per-claim verdicts with individual PubMed citations would make the system significantly more useful for detailed fact-checking workflows, where researchers need to know not just whether a video is problematic but which specific statements require correction.

7.2 Alignment with WHO Priorities

A 2025 WHO-commissioned Delphi study on responsible AI in public health, drawing on input from 54 experts across 27 countries, identified social listening and trend analysis as the top AI opportunities for infodemic management, and ranked equity and transparency as the leading responsible-AI principles (Mahl et al., 2025). The WHO Infodemic Monitor directly addresses the transparency requirement: every classification comes with a natural language explanation, specific transcript evidence, and peer-reviewed citations. The claim-level work in Task 5 takes this further by linking decisions to individual statements rather than entire videos.

The equity principle, ensuring AI tools work reliably across demographic groups and content types, points to a known limitation of the current system. The PubMed grounding and Whisper transcription pipelines are optimized for English-language content. Expanding language coverage is a necessary step before the system could be deployed globally in a WHO context.

7.3 Limitations of the Current System

Several limitations are worth stating directly. First, the production system's validation was conducted on four videos, which is a meaningful proof of concept but not a statistically rigorous evaluation. A formal evaluation against a labeled benchmark would establish measurable performance baselines. Second, the system currently operates on downloaded video files. Supporting direct URL ingestion from TikTok, Reels, and Shorts would dramatically lower the barrier for researchers. Third, the Celery worker is currently hosted locally rather than fully in the cloud. Migrating it to Railway's Hobby plan resolves this. Fourth, Whisper has documented hallucination risks in approximately 1% of audio samples. Voice-activity-detection pre-segmentation and per-segment confidence filtering are known mitigations.

8. Future Work

The following priorities are identified for continued development:

Full Cloud Deployment

Moving the Celery analysis worker from local operation to Railway's cloud infrastructure is the immediate next step. This makes the system available around the clock without requiring any team member's machine to be running.

Direct URL Ingestion

Rather than requiring researchers to download and re-upload videos, the system should accept a TikTok, Reels, or Shorts URL directly and handle video retrieval internally. This is the most significant usability improvement for field researchers.

Stronger PubMed Evidence Search

The current grounding step uses keyword-based queries generated by the LLM. Replacing this with MeSH term queries and hybrid BM25 and dense retrieval would significantly improve the relevance and recall of retrieved citations.

Confidence Calibration and Evaluation Harness

The system currently returns confidence scores based on LLM self-assessment, which is not well calibrated. Building an evaluation harness against a labeled benchmark with precision, recall, and F1 measured per label would establish objective performance baselines.

OCR Re-enable

Based on Task 3's findings, OCR should be re-enabled with a confidence-threshold trigger rather than always-on. This provides the accuracy benefit of multimodal processing in cases where it matters, without paying the cost in cases where it does not.

Three-way Ensemble Routing

Task 4's model agreement analysis provides the foundation for an ensemble routing layer. Running Llama 2, Mistral, and Gemma on low-confidence cases and routing based on inter-model agreement would provide a more robust secondary verification step.

Claim-Level Classification Integration

Completing the claim-level work from Task 5 with a labeled evaluation dataset, per-claim PubMed grounding, and a UI drill-down view would substantially improve the system's utility for detailed fact-checking workflows.

Language Coverage

Extending transcription and classification support to Spanish, Portuguese, Hindi, and Arabic is essential for global deployment in a WHO context.

9. Conclusion

The WHO Infodemic Monitor demonstrates that a small team of graduate and undergraduate students can build a production-grade, cloud-deployed AI system for health misinformation detection that is meaningfully more capable than the research prototypes that preceded it. The system moved from a locally-run Streamlit prototype requiring a dedicated GPU to a fully cloud-hosted platform accessible from any browser, processing real health videos in under 10 seconds with nuanced, evidence-grounded explanations.

The three research tracks contributed findings that directly inform the system's next iteration. Audio is sufficient for most short-form health video classification, and OCR should be triggered conditionally. Open-weight LLMs agree on clear cases in ways that can be used as production routing signals. Claim-level outputs offer meaningfully better transparency than video-level labels.

Health misinformation in short-form video is not a problem that AI alone can solve. Platform accountability, media literacy, and regulatory frameworks all have roles to play. A tool that gives public health researchers fast, explainable, evidence-grounded analysis of viral health content, without requiring manual review of thousands of videos, is a meaningful contribution to the infodemic management toolkit that WHO and its partners are actively building.

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